



Amazon S3 — Complete Notes

Simple Storage Service · Object storage on AWS

AWS S3

Durability 11 nines

Storage Object

Max Object 5 TB

A reader-friendly, all-in-one reference for Amazon S3.

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1. What is S3

Amazon S3 is an **object storage** service from AWS. It stores data as **objects** inside containers called **buckets**. Designed for **99.999999999% (11 nines)** durability and **99.99%** availability, with virtually unlimited capacity — you never provision disks. You pay only for what you store, the requests you make, and data transferred out.

💡 **Important:** S3 is *not* a file system and *not* block storage. It is flat object storage accessed over HTTP/HTTPS via a REST API. "Folders" are just a naming convention using `/` in the key.

Common uses: backups, static website hosting, data lakes, media storage, software distribution, log storage, big-data analytics, archival.

2. 🧩 Core Concepts

Concept	Description
Bucket	Top-level container for objects. Name is globally unique across all AWS accounts. Lives in one region .
Object	The data + metadata. Made of Key, Value, Version ID, Metadata, ACL info. Size 0 B → 5 TB .
Key	Full unique name of an object, e.g. <code>images/2024/photo.jpg</code> .
Prefix	Start of a key, used to group/filter objects (simulates folders).
Delimiter	Usually <code>/</code> , used by list ops to roll keys into folder-like groups.
Region	Geographic location where the bucket physically lives — chosen for latency, cost, compliance.

Bucket naming rules: 3–63 chars · lowercase, numbers, hyphens, dots · must start/end with letter or number · no uppercase, no underscores · cannot look like an IP address.

Upload limits: single `PUT` max **5 GB** → above that you **must** use multipart upload (recommended for anything > 100 MB).

3. 🗄️ Storage Classes

All classes offer the same **11 nines** durability (*except One Zone, which is single-AZ*). They trade off cost vs. access speed vs. retrieval fees.

Class	Best for	Access speed	Min. duration	Notes
Standard	Frequently accessed data	Milliseconds	—	Default. ≥3 AZs. No retrieval fee.
Intelligent-Tiering	Unknown / changing patterns	Milliseconds	—	Auto-moves between tiers. Small monitoring fee, no retrieval fee.
Standard-IA	Infrequent, rapid access	Milliseconds	30 days	Cheaper storage + per-GB retrieval fee.
One Zone-IA	Re-creatable / secondary data	Milliseconds	30 days	Single AZ → ~20% cheaper, lower resilience.
Glacier Instant Retrieval	Archive, instant access	Milliseconds	90 days	Cheap storage, higher retrieval cost.
Glacier Flexible Retrieval	Backups / DR archives	Minutes–hours	90 days	Expedited (1–5 min), Standard (3–5 h), Bulk (5–12 h).
Glacier Deep Archive	Long-term compliance	12–48 hours	180 days	Lowest cost in AWS.
S3 on Outposts	On-prem data residency	Local	—	S3 on AWS Outposts hardware.

4. Durability, Availability, Consistency

- **Durability** — **99.999999999% (11 nines)**. Store 10M objects and expect to lose ~1 every 10,000 years. Achieved via redundant storage across devices and (for most classes) multiple AZs.
- **Availability** — **varies**: Standard **99.99%**, Standard-IA **99.9%**, One Zone-IA **99.5%**.
- **Consistency** — **strong read-after-write** for all operations (since Dec 2020). A read right after a write/overwrite/delete returns the latest data.  No more eventual-consistency surprises.

5. Versioning

- Keeps **multiple versions** of an object in the same bucket — protects against accidental deletes/overwrites.
- States: **Unversioned** (default) · **Enabled** · **Suspended**. Once enabled it can only be *suspended*, never fully disabled.
- Each version gets a unique **Version ID** (`null` for pre-versioning objects).
- A **DELETE** adds a **delete marker** — it does *not* erase data; the prior version stays recoverable. To truly delete, delete a specific version ID.
- Increases storage cost (every version is stored) → pair with **lifecycle rules** to expire old versions.
- **MFA Delete**: optional protection requiring an MFA token to delete versions or change versioning state (root-only, via CLI).

6. 🔄 Lifecycle Management

Rules that automatically **transition** or **expire** objects to save cost.

- **Transition actions** — move to a cheaper class after N days (e.g. *Standard* → *Standard-IA* after 30 days → *Glacier* after 90 days).
- **Expiration actions** — delete objects after N days, delete old versions, abort incomplete multipart uploads, remove expired delete markers.

Rules can target the whole bucket, a prefix, or object **tags**. Great for log retention, compliance, and cost optimization.

⚠️ Cannot transition to IA before 30 days; very small objects may not be worth moving.

7. 🛡️ Security & Access Control

Everything in S3 is **private by default**. Only the bucket owner has access.

Layer	What it does
IAM Policies	Identity-based — what <i>your</i> users/roles can do.
Bucket Policies	JSON resource policy on a bucket; cross-account access, enforce HTTPS/encryption, restrict by IP. Most common bucket-wide control.
ACLs	Legacy object/bucket grants. AWS recommends disabling them ("Bucket owner enforced").
Block Public Access	Account/bucket safety switch (4 settings). ON by default — prevents accidental public exposure.
Access Points	Named endpoints with their own policies, for managing shared buckets/data lakes at scale.
Presigned URLs	Time-limited URL granting access to one object without making it public.
VPC Endpoint	Private access to S3 from a VPC without going over the public internet.

8. 🔑 Encryption

In transit: use HTTPS/TLS — can be enforced via bucket policy (`aws:SecureTransport`).

At rest (server-side encryption, SSE):

Type	Who manages keys	Notes
SSE-S3 (AES-256)	AWS / S3	Default since Jan 2023 — all new objects encrypted automatically.
SSE-KMS	AWS KMS	Audit trail (CloudTrail), key rotation, fine-grained access. Slight cost + API limits.
SSE-C	Customer (per request)	AWS doesn't store the key.
Client-side	You, before upload	AWS never sees plaintext.

DSSE-KMS: dual-layer KMS encryption for high-compliance needs.

9. 🚀 Data Transfer & Performance

- **Multipart Upload** — split a large object into parallel parts. Required > 5 GB, recommended > 100 MB. Up to **10,000 parts**, each ≤ 5 GB. Resumable. *Abort incomplete uploads via lifecycle rule to avoid orphan charges.*
- **Transfer Acceleration** — uses CloudFront edge locations + AWS backbone to speed long-distance transfers (extra cost).
- **Byte-Range Fetches** — download just part of an object via the **Range** header; enables parallelism.
- **Performance** — scales to **≥ 3,500 writes** and **≥ 5,500 reads per second per prefix**. Spread keys across prefixes for more parallel throughput.
- **S3 Select** — pull a subset of data using SQL (CSV/JSON/Parquet) so you transfer less.

10. 🌐 Replication

Automatically copy objects to another bucket. **Requires versioning** on source *and* destination.

Type	Scope	Use cases
CRR (Cross-Region)	Different region	Compliance, lower latency for distant users, disaster recovery.
SRR (Same-Region)	Same region	Log aggregation, prod/test sync, cross-account.

- Can replicate to a different **account** and a different **storage class**.
- Only replicates **new** objects after the rule is enabled (use **Batch Replication** for existing ones).
- **Replication Time Control (RTC)** offers a 15-minute SLA.

11. 🌐 Static Website Hosting

- Host static sites (HTML/CSS/JS/images) — **no server-side code**.
- Enable *Static website hosting*, set index + error documents.
- Bucket must allow public read (or front it with **CloudFront**).
- Endpoint: `bucket-name.s3-website-region.amazonaws.com`
- For HTTPS + custom domain + caching → put **CloudFront** in front.

12. 🛎 Event Notifications & Integration

- S3 emits events on actions (object created, deleted, restored...).
- **Targets:** SNS (notify), SQS (queue), Lambda (run code), EventBridge (advanced routing/filtering).
- **Common pattern:** upload object → trigger Lambda → process file (resize image, ingest data, etc.).

13. 📊 Monitoring, Logging, Analytics

Tool	Purpose
Server Access Logging	Detailed request records, stored in another bucket.
CloudTrail	Audit S3 API calls (management + data events).
CloudWatch	Metrics: bucket size, object count, request metrics.
Storage Lens	Org-wide usage/activity visibility + cost recommendations.
S3 Inventory	Scheduled report (CSV/ORC/Parquet) of objects + metadata.
Storage Class Analysis	Analyzes access patterns to recommend moving to IA.

14. 🛒 Other Features

- **Object Lock (WORM)** — Write Once Read Many. Prevents delete/overwrite for a retention period or legal hold. Modes: **Governance** (override with permission) and **Compliance** (no one, not even root). Requires versioning.
- **Object Tags** — up to 10 key-value tags per object; drive lifecycle, access control, cost allocation, filtering.
- **Requester Pays** — the requester (not owner) pays for requests + transfer; for sharing large datasets.
- **S3 Batch Operations** — bulk actions on billions of objects (copy, tag, restore, invoke Lambda).
- **S3 Object Lambda** — transform data on retrieval (redact, resize, convert) without storing copies.
- **Mountpoint for S3** — mount a bucket as a local filesystem for read-heavy workloads.

15. 💰 Pricing

You pay for:

1. **Storage** — per GB/month, by class.
2. **Requests** — per 1,000 (PUT, GET, LIST...), varies by class.
3. **Data transfer OUT** — to the internet (*IN is free; S3 ↔ EC2 same-region is free*).
4. **Retrieval fees** — for IA & Glacier classes (per GB retrieved).
5. **Management features** — Intelligent-Tiering monitoring, inventory, analytics, replication.

6. **Early-delete fees** — deleting before a class's minimum duration.

 **Cost tips:** use lifecycle rules → cheaper classes · Intelligent-Tiering for unknown patterns · delete old versions + incomplete multipart uploads · keep compute in the same region · use a VPC Gateway Endpoint to dodge NAT/transfer charges.

16. Common CLI Commands

```
aws s3 mb s3://my-bucket           # make/create bucket
aws s3 rb s3://my-bucket           # remove bucket
aws s3 ls                           # list buckets
aws s3 ls s3://my-bucket/          # list objects
aws s3 cp file.txt s3://my-bucket/  # upload a file
aws s3 cp s3://my-bucket/file.txt . # download a file
aws s3 sync ./localdir s3://my-bucket/ # sync a folder
aws s3 rm s3://my-bucket/file.txt   # delete an object
aws s3 presign s3://my-bucket/file.txt # generate presigned URL
```

Low-level API uses `aws s3api ...` for fine control (versioning, policies, multipart, etc.).

17. Key Limits & Numbers

Item	Value
Max object size	5 TB
Max single PUT upload	5 GB (multipart above this)
Max multipart parts	10,000 (each ≤ 5 GB)
Bucket name	3–63 chars, globally unique, lowercase
Buckets per account	100 default (up to 1,000 on request)
Durability	11 nines (99.99999999%)
Standard availability	99.99%
Object tags	10 max per object
User-defined metadata	2 KB limit
Request rate per prefix	3,500 writes / 5,500 reads per second
Min storage duration	IA 30d · Glacier 90d · Deep Archive 180d

18. 🚧 Ways to Connect to S3

Everything ultimately calls the same REST API — only the entry point differs. Almost every method needs valid **credentials + IAM permissions**.

A. Credentials first (authentication)

- **IAM user access keys** — Access Key ID + Secret. Used by CLI/SDK. Avoid long-lived keys.
- **IAM roles** (*preferred*) — EC2 instance role, Lambda exec role, ECS task role, EKS IRSA. Temporary creds, nothing stored in code.
- **STS temporary credentials** — short-lived tokens via AssumeRole / federation / web identity. Best for cross-account / temporary access.
- **Root account** — never for normal access.

B. AWS Management Console (web UI)

Browser-based — create buckets, upload/download, set policies. Good for manual/admin tasks.

C. AWS CLI

`aws s3 ...` (high-level) and `aws s3api ...` (low-level). Configure with `aws configure` or named profiles (`--profile myprofile`). Great for scripts/automation.

D. AWS SDKs (in code)

Python (**boto3**), JS/Node (**@aws-sdk/client-s3**), Java, **Go** (**aws-sdk-go-v2**), .NET, Ruby, PHP, C++, Rust... SDK handles SigV4 signing, retries, multipart, pagination.

```
# Python (boto3)
import boto3
s3 = boto3.client("s3")
s3.upload_file("local.txt", "my-bucket", "key.txt")
s3.download_file("my-bucket", "key.txt", "local.txt")
```

```
// Go (aws-sdk-go-v2)
cfg, _ := config.LoadDefaultConfig(ctx)
client := s3.NewFromConfig(cfg)
client.PutObject(ctx, &s3.PutObjectInput{ /* ... */ })
```

E. REST API directly (HTTP)

Raw HTTPS requests signed with **AWS Signature v4**.

```
https://my-bucket.s3.region.amazonaws.com/key      (virtual-hosted)
https://s3.region.amazonaws.com/my-bucket/key     (path-style, legacy)
```

Usually let an SDK/CLI do the signing.

F. Presigned URLs

Time-limited signed URL to GET or PUT a specific object **without** AWS credentials. Common for browser uploads/temporary downloads.


```
aws s3 presign s3://bucket/key --expires-in 3600
```

G. Mount as a filesystem

- **Mountpoint for Amazon S3** — official AWS client; mounts a bucket as a folder (read-heavy / sequential).
- **s3fs-fuse** — third-party FUSE mount (slower, not a true FS).
- **AWS Storage Gateway (File Gateway)** — S3 via NFS/SMB with local caching for on-prem apps.

H. Private network connectivity

- **VPC Gateway Endpoint** — reach S3 privately from a VPC, no IGW/NAT, no data leaves AWS. **Free.**
- **VPC Interface Endpoint (PrivateLink)** — private IP in your subnet (has cost; on-prem-to-VPC).
- **Direct Connect / VPN** — connect on-prem networks to AWS, then reach S3.

I. Third-party / GUI tools

Cyberduck, S3 Browser, WinSCP, Rclone, Transmit — speak the S3 API using your keys. Good for manual transfers and cross-cloud sync (rclone).

J. Through other AWS services (S3 as backend)

CloudFront (CDN), **Athena** (SQL over S3), **Glue**, **EMR/Spark**, **Redshift Spectrum**, **Lambda** triggers, **DataSync** (bulk migration), **Transfer Family** (SFTP/FTPS/FTP front end).

K. S3-compatible access

The S3 API is a de-facto standard — point the SDK/CLI at a custom **endpoint URL** to talk to MinIO, Wasabi, Ceph, etc.

Quick pick guide

Need	Use
One-off manual task	Console or GUI tool
Scripting / automation	AWS CLI
Application code	AWS SDK (boto3, aws-sdk...)
Let a user upload/download	Presigned URL
App inside a VPC	IAM role + VPC Gateway Endpoint
Legacy app needs a filesystem	Mountpoint / Storage Gateway
Serve files to the web	CloudFront in front of S3

19. Quick Mental Model

- **S3 = unlimited buckets of objects**, accessed over HTTP, extremely durable.
- Pick a **storage class** by how often you read data and how fast you need it back.
- Everything is **private by default** — open access deliberately via IAM + bucket policy; keep **Block Public Access ON**.

- Turn on **Versioning + Lifecycle** rules to stay safe and cheap.
 - **Encrypt** in transit (HTTPS) and at rest (SSE on by default).
 - Use **Multipart** for big files, **Transfer Acceleration / CloudFront** for speed.
 - Use **Events + Lambda** to make storage reactive.
 - Use **Replication** for DR/compliance, **Object Lock** for WORM compliance.
 - **Monitor** with CloudWatch, CloudTrail, Storage Lens; optimize cost relentlessly.
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20. ? Common Interview Questions

Rapid-fire questions interviewers ask about S3:

- **Q: What is S3's consistency model?** — Strong read-after-write consistency for **all** operations (PUT, overwrite, DELETE) since Dec 2020. A read right after a write returns the latest data.
 - **Q: S3 vs EBS vs EFS?** — S3 = object storage over HTTP (no filesystem). EBS = block storage for a single EC2 (one AZ). EFS = shared NFS filesystem across many instances.
 - **Q: How do you secure an S3 bucket?** — Block Public Access ON, least-privilege IAM + bucket policy, default SSE encryption, enforce HTTPS (`aws:SecureTransport`), VPC endpoint for private access, enable versioning + logging.
 - **Q: Why use versioning?** — Protects against accidental delete/overwrite; a DELETE adds a *delete marker* (recoverable) rather than erasing data.
 - **Q: Object size limits?** — Up to 5 TB; single PUT max 5 GB → use **multipart upload** above that (recommended > 100 MB).
 - **Q: When Glacier vs Standard-IA?** — IA for infrequent but instant access; Glacier (Instant/Flexible/Deep Archive) for archival where retrieval time/cost is acceptable.
 - **Q: What is a pre-signed URL?** — A time-limited signed URL granting temporary access to one object without making it public.
 - **Q: How to reduce S3 cost?** — Lifecycle rules to cheaper classes, Intelligent-Tiering for unknown patterns, delete old versions + incomplete multipart uploads, keep compute in-region.
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 Notes compiled as a quick reference for Amazon S3.